

Iteration 2 Testing Plan

Project: AI-Assisted Social Media Bot Risk Scoring Tool **Primary audience:** External Social Media Safety Analyst **Scope:** Iteration 2 hardening of the 5-Feature, 12-User-Story MVP **Document purpose:** Direct, browser-based verification plus existing automated checks **Code baseline:** the submitted commit on `fix/iteration2-ui-audit-findings` (record its SHA before execution) **Working branch:** `fix/iteration2-ui-audit-findings` **Document status:** Ready for team/tutor execution; selected walkthrough results are recorded in `docs/test_results.md` **Last updated:** 9 August 2026

A written test is not a passed test. The executor must record the actual visible result and Pass/Fail. Screenshots, HAR files, Console exports and shared-folder links are not required for this Testing Plan.

1. Purpose and Scope

This plan verifies the visible Iteration 2 MVP as a tutor or target-user proxy would use it: open the local website, select or enter a sample account, click the controls, upload tracked CSV examples, and compare the page with an explicit expected result.

The application is an offline human-review aid. It must not contact live Twitter/X, make a final bot/human determination, or perform moderation, suspension, reporting or another platform action.

The plan covers:

- direct GUI interaction for intake, preview, completeness, scoring, explanation, model information, decisions, follow-up, batch and filters;
- keyboard, focus, text/colour, responsive layout and 200% zoom;
- invalid, unknown, Insufficient and stopped-local-service states;
- existing pytest, JavaScript syntax and Run Sheet checks; and
- a simple truthful execution record without mandatory media attachments.

2. Test Environment and Setup

2.1 Preconditions

- Work from the repository root.
- Use the project Conda environment `fit5238-backend`.
- Use the tracked offline demonstration records and model artifact.
- Use only the tracked project-dataset samples or synthetic boundary fixtures; do not add live/current account data.
- Test current Chrome and Edge and record their exact versions.
- Use a clean browser tab at `http://127.0.0.1:8000/`.

2.2 Start the application

```
conda activate fit5238-backend
python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8000
```

Open `http://127.0.0.1:8000/`. Stop the service with `Ctrl+C`.

2.3 Minimum execution record

For every WEB-EX case record:

Field	Required value
Test ID	WEB-EX-01 to WEB-EX-18
Executor and date	Name, date and timezone
Build	Branch and commit SHA or working-tree note
Browser	Chrome/Edge name and exact version
Display	Viewport and zoom
Input/actions	Exact account/file and controls used
Actual result	What was visibly observed

Field	Required value
Status	Pass, Fail, Blocked or Not executed
Defect	Bug ID and retest result when failed

No screenshot, HAR, Console export or shared-folder link is required.

3. Tracked Sample Data

3.1 Sample account catalogue

Browser input	Scenario	Expected visible baseline
20611469, @DeFotis, or the Low selector option	Static row from Twitter Human Bots	@DeFotis; 100% completeness; 2 / 100 Low
396039913, @OGLexa, or the Medium selector option	Static row from Twitter Human Bots	@OGLexa; 100% completeness; 30 / 100 Medium
2250581388, @everyletterbot, or the High selector option	Static row from Twitter Bot Training Data 2	@everyletterbot; 100% completeness; 91 / 100 High; missing location shown
demo_exact_50	Inclusive completeness boundary	@half_evidence; exactly 50%; eligible; caveat remains; 53 / 100 Medium
demo_incomplete_01	Insufficient evidence	@limited_record; 25%; Insufficient data; no score or risk band
bad_identifier!	Invalid local input	Adjacent validation message; assessment does not start
not_in_offline_fixture	Valid format but unavailable offline	Controlled unavailable-account message

The first three rows are copied from the two tracked project datasets without their training label. Dataset names show provenance only; the interface does not present a bot/human label. The previous demo_low_01, demo_medium_01 and demo_high_01 fixtures remain after these rows for regression and batch-file compatibility, but they are no longer the landing-page selector examples.

The numeric expectations are version-bound to xgb-offline-v1, threshold-v1 and data/fixtures/demo_accounts.csv. Re-baseline them if the model, thresholds or fixture changes.

3.2 Browser-upload files

File	Purpose and expected result
tests/fixtures/browser_batch.csv	Seven mixed rows: 4 Completed and 3 Failed
tests/fixtures/browser_batch_empty.csv	Header only: accepted no-op with 0 total, 0 completed, 0 failed
tests/fixtures/browser_bad_header.csv	Wrong header: controlled error naming required account_id
tests/fixtures/browser_not_csv.txt	Wrong type: controlled file-type error
tests/fixtures/browser_batch_100_rows.csv	Exact limit accepted; 100 unknown rows reported independently
tests/fixtures/browser_batch_101_rows.csv	Limit exceeded; controlled error and no partial table
Local-only browser_batch_over_100kb.csv generated in WEB-EX-13	Browser rejects the file before reading or submitting it

4. Hands-on Web Interaction Examples

WEB-EX-01 - Landing page and offline purpose

Maps to: global product rules, US1.1, US4.2

1. Start the application and open the home page.
2. Read the header, offline badge, human-oversight notice and form labels.
3. Confirm single-account and batch sections are available.

Expected visible result: the first view says Review Twitter/X accounts for possible bot activity. and explains the score, risk level, contributing factors and data-quality warnings. It states that final judgement remains with the user and no account is reported, suspended or moderated. The account input, demonstration selector, Begin assessment control, CSV control and the permanent **Understand this review tool** link are labelled. No wording promises live Twitter/X access or automated enforcement. The technical **Developer API** link is available in the footer without competing with the account-intake controls.

Status: Not executed.

WEB-EX-02 - Invalid and blank account input

Maps to: US1.1 AC2/AC6

1. Submit a blank identifier.
2. Enter bad identifier! and select **Begin assessment**.
3. Correct the value by typing @DeFotis.

Expected visible result: blank and malformed values show an adjacent, actionable message and do not start an assessment. The invalid state clears when the value is corrected. The identifier remains reachable and labelled.

Status: Not executed.

WEB-EX-03 - Low result and source/model separation

Maps to: US1.1, US1.2, US2.1, US2.2, US3.1

1. Select the Low demonstration scenario or enter @DeFotis.
2. Select **Begin assessment**.
3. Inspect the Source data, completeness and Model-generated output sections.

Expected visible result: the preview identifies @DeFotis and groups profile, activity and network fields. Completeness is 100%. The result is **2 / 100 Low** and uses the word Low as well as colour. It is explicitly an out-of-100 review score, not a probability percentage. The page moves to the new result and shows a plain-language suggested human-review priority. Model version, threshold version, assessment time, uncertainty, no-verdict wording and up to three factor rows are visible. The source badge says **Offline demonstration record**. Every factor uses a plain label, direction and observed value; encoded Boolean values appear as Yes/No rather than 1/0.

Status: Not executed.

WEB-EX-04 - Start new and Medium result in the same tab

Maps to: US1.3, US2.2

1. Complete WEB-EX-03.
2. Select **Start new assessment**.
3. Run @OGLexa.

Expected visible result: the previous Low preview, score, factors, decision and follow-up state clear. Focus returns to the first assessment control. The new preview identifies @OGLexa, completeness is 100%, and the result is 30 / 100 Medium with no stale Low content.

Status: Not executed.

WEB-EX-05 - High result, missing source values and factors

Maps to: US1.2, US2.2, US3.1, US3.2

1. Run @everyletterbot.
2. Compare the source preview with the model output.
3. Read the factor rows from top to bottom.
4. Open **Review model information and limitations**.

Expected visible result: the preview identifies @everyletterbot and shows its unavailable location as Not available rather than inventing a value. The result is 91 / 100 High. No more than three factors appear strongest-first with direction and observed value. The uncertainty and Triage evidence, not a verdict. statement remain visible. A direct link opens model information.

Status: Not executed.

WEB-EX-06 - Exactly 50% completeness boundary

Maps to: US2.1 AC1/AC3, US2.2

1. Enter `demo_exact_50`.
2. Select **Begin assessment**.
3. Compare completeness, missing-feature caveat and result.

Expected visible result: completeness is exactly 50% and remains eligible. Missing features and the caveat stay visible. The model result is 53 / 100 Medium. This confirms that the 50% boundary is inclusive.

Status: Not executed.

WEB-EX-07 - Insufficient evidence without a risk band

Maps to: US2.1 AC2/AC4, US3.2 AC4, US4.1 AC1, US4.3 AC1

1. Enter `demo_incomplete_01`.
2. Select **Begin assessment**.
3. Inspect completeness, result, decision, follow-up and model-information areas.

Expected visible result: completeness is 25%, status is Insufficient data, and missing features use plain labels. No score, Low/Medium/High band or Confirm/Override form appears. Follow-up and model-information access remain available.

Status: Not executed.

WEB-EX-08 - Successful result followed by unknown account

Maps to: US1.3 AC1-AC4, US2.2 AC7

1. Complete WEB-EX-05.
2. Without selecting Start new, replace the identifier with `not_in_offline_fixture`.
3. Submit again.

Expected visible result: a controlled unavailable-account message appears. The old preview, completeness, 91 / 100 score, factors, decision and follow-up panels are no longer actionable. Start new remains available. A valid-format unknown identifier is not described as malformed input.

Status: Not executed.

WEB-EX-09 - Stopped local service and Retry

Maps to: US1.3 AC1-AC3/AC6, US2.2 AC7

1. Keep the web page open and stop Uvicorn.
2. Submit `@DeFot.is`.
3. Confirm the error contains no partial or old score.
4. Restart Uvicorn and select **Retry** once.

Expected visible result: a plain-language local-service error offers recovery. After restart, Retry reuses `@DeFot.is` and produces one Low 2 / 100 result. No duplicate decision or follow-up state appears.

Status: Not executed.

WEB-EX-10 - Confirm/Override reason and duplicate prevention

Maps to: US4.1 AC1-AC5

1. Complete WEB-EX-05.
2. Select **Override suggested review priority** and try an empty reason.
3. Enter `manual browser review` and select **Record decision** twice rapidly.
4. In a fresh Medium assessment, record Confirm once.

Expected visible result: empty Override is blocked. A valid decision produces one acknowledgement that the human decision is final and no platform action was taken. The submit control prevents a second decision for the same assessment. Confirm does not require an override reason. Both stored-note areas visibly warn the reviewer not to enter private or sensitive information such as passwords, private messages, phone numbers or email addresses.

Status: Not executed.

WEB-EX-11 - Follow-up add, update, clear and reset

Maps to: US4.3 AC1-AC5, US1.3 AC4-AC6

1. Complete @0GLexa.
2. Save a flag with initial human review.
3. Open Model Information, return, and confirm the flag reason/status remains.
4. Change it to updated human review and save again.
5. Select **Clear flag**.
6. Start new, run demo_incomplete_01, save and clear another flag.

Expected visible result: status changes through Flagged, updated Flagged and Cleared without duplicate visible state. Every acknowledgement states that no platform action occurred. Insufficient data supports follow-up but never shows Confirm/Override. Return navigation preserves the visible flag and reason; Start new clears the old reason/status.

Status: Not executed.

WEB-EX-12 - Mixed batch partial success

Maps to: US5.1 AC1-AC4

1. Select tests/fixtures/browser_batch.csv.
2. Confirm the English filename replaces No file selected.
3. Select **Assess batch offline** and observe the busy state.
4. Inspect all result rows.

Expected visible result: while processing, the button and file picker are disabled, the offline-processing label is visible, counts start at 0/0 and an old table is hidden. Completion shows 7 total, 4 Completed and 3 Failed. Low 2 / 100, Medium 30 / 100, High 91 / 100 and Insufficient rows complete. Invalid, duplicate and unknown rows fail independently. No platform action is claimed.

Status: Not executed.

WEB-EX-13 - File size, type, header and 0/100/101 boundaries

Maps to: US5.1 AC1/AC2

1. Submit each tracked file from section 3.2 in order. If the picker filters out TXT, select All files before choosing browser_not_csv.txt.
2. Create one local-only size-boundary file, then select it in the browser:

```
$oversizeCsv = Join-Path $env:TEMP "browser_batch_over_100kb.csv"
$content = "account_id`n" + ("x" * 100001)
Set-Content -LiteralPath $oversizeCsv -Value $content -NoNewline -Encoding ascii
```

Expected visible result: wrong type, oversize file and wrong header show controlled errors; the empty file shows 0/0/0; the 100-row file is accepted and reports all 100 unknown rows; the 101-row file is rejected without a partial results table. The busy state clears after every attempt and the visible filename matches the file chosen. The oversize message names the 100 KB limit and no processing request starts.

Status: Not executed.

WEB-EX-14 - Sort, filter, matching count and clear

Maps to: US5.2 AC1-AC4

1. Complete WEB-EX-12.
2. Sort Highest first, then Lowest first.
3. Test High, Insufficient and Not reviewable separately.
4. Combine High + Eligible + Unreviewed.
5. Select **Clear filters**.

Expected visible result: scored order is 91/30/2 or 2/30/91 (each out of 100) and unscored rows remain last. Matching counts are High 1, Insufficient 1, Not reviewable 3 and combined 1. Active filters are described. Clear restores the original seven rows and values without showing a new processing run.

Status: Not executed.

WEB-EX-15 - New batch clears stale filters and results

Maps to: US5.1, US5.2, integration regression

1. Leave a High filter active after WEB-EX-12.
2. Select and submit another tracked CSV.
3. Observe controls before and after processing.

Expected visible result: filter controls reset before the new batch runs, old results hide, counts reset and the final table contains only the new batch. The summary reports no active filters and the correct new count.

Status: Not executed.

WEB-EX-16 - Complete model-information visual review

Maps to: US3.2 AC1-AC4, tutor usability feedback

1. From the initial home-page view, open **Understand this review tool** and confirm the ordinary-language guidance is visible before technical details.
2. Return, run @everyLetterbot, then open model information from the scored result.
3. Read **How to use this page** and follow its three links to **What this tool checks**, **How to read the result**, and **When the result needs extra care**.
4. Confirm **Technical model and evaluation record** is collapsed by default, then expand it and read all twelve public-data feature entries.
5. Compare the plain-language risk-score guidance with the recorded technical F1, precision, recall and calibration evidence.
6. Select **Return to assessment**.

Expected visible result: the page first explains the tool without requiring machine-learning knowledge and says the reference does not change the current assessment. Technical terms and exact evidence are available on demand rather than dominating the first view. The expanded record reports xgb-offline-v1, threshold-v1, both evaluation files, evaluation date, F1 0.5859, precision 0.4158, recall 0.9915 and validation mean cost 0.266149. It names excluded graph evidence, cross-dataset limits, prevalence shift, concept drift, false-positive risk and prohibited automated enforcement. The Profile completeness definition distinguishes that single model input from the overall 50% scoring gate. Return restores the same @everyLetterbot 91 / 100 High assessment and moves back to its result; the score, band, factors and assessment time do not disappear or change.

Status: Not executed.

WEB-EX-17 - Keyboard, focus and text alternatives

Maps to: US1.1 AC6, US2.2 AC5, US5.2 controls

1. Use only Tab, Shift+Tab, arrow keys, Enter and Space.
2. Activate the Skip link.
3. Run Low, record a decision, save/clear follow-up, upload a batch, change each filter and clear filters.
4. Repeat invalid input and Start new.

Expected visible result: every control is reachable, labelled and has visible focus; focus is not trapped; the Skip link moves to main content; validation and Start new return focus appropriately; risk meaning is available in text and is not colour-only.

Status: Not executed.

WEB-EX-18 - Chrome/Edge viewport and 200% zoom matrix

Maps to: accessibility and all visible workflows

Repeat the @DeFotis Low, @everyLetterbot High, Insufficient and WEB-EX-12 paths in current Chrome and Edge at:

Viewport	Zoom
1280 x 720	100% and 200%
768 x 1024	100% and 200%
375 x 667	100% and 200% where the browser supports it

Expected visible result: headings, notices, source values, factors, errors and controls do not overlap or clip. Long text wraps. The single-account page does not require horizontal page scrolling. The batch table scrolls inside its labelled region while filter summary and Clear filters remain reachable.

Status: Not executed.

5. Existing Automated and Operational Checks

Run these after the documentation/sample-file change:

```
python -m pytest --basetemp ".pytest-tmp"
node --check frontend/app.js
node --check frontend/api-client.js
node --check frontend/assessment-state.js
node --check frontend/assessment-view.js
node --check frontend/batch-results.js
node --check frontend/batch-controller.js
node --check frontend/review-controller.js
python -m scripts.verify_run_sheet
python -m pip check
git diff --check
```

Expected: all 91 currently collected tests pass, all seven JavaScript files parse, the Run Sheet prints its PASS line, dependencies are consistent, and Git reports no whitespace errors.

The tracked GitHub Actions and GitLab CI configurations run the same pytest, JavaScript, Run Sheet and pip check gates after repository delivery. A local pass remains the evidence for this document; a remote CI badge is not assumed.

The existing automated suite covers data cleaning, feature mapping, exact completeness logic, deterministic scoring, error contracts, privacy, decisions/follow-up, batch processing, filters and cross-story integration. Source/model timeout and malformed-model states are not production GUI buttons; they remain automated fault-injection cases rather than teacher click examples.

6. Acceptance-Criterion Traceability

User Story	Direct browser evidence	Supporting automated evidence
US1.1 Intake	WEB-EX-01 to 05, 17	tests/test_us1_1_account_intake.py
US1.2 Preview	WEB-EX-03 to 05	tests/test_us1_2_account_preview.py
US1.3 Recovery/reset	WEB-EX-04, 08, 09, 11, 15, 17	tests/test_us1_3_recovery_reset.py
US2.1 Completeness	WEB-EX-03, 06, 07	tests/test_us2_1_completeness.py, tests/test_features.py
US2.2 Risk result	WEB-EX-03 to 09, 17, 18	tests/test_us2_2_risk_scoring.py, tests/test_model_service.py
US3.1 Explanation	WEB-EX-03, 05, 18	tests/test_us3_1_explanation_uncertainty.py
US3.2 Model information	WEB-EX-05, 07, 16	tests/test_us3_2_model_information.py
US4.1 Decision	WEB-EX-07, 10	tests/test_us4_1_decision.py
US4.2 Data protection	WEB-EX-01, 10, 11	tests/test_us4_2_data_protection.py
US4.3 Follow-up	WEB-EX-07, 11	tests/test_us4_3_follow_up.py
US5.1 Batch	WEB-EX-12, 13, 15, 18	tests/test_us5_1_batch_assessment.py
US5.2 Sort/filter	WEB-EX-14, 15, 17, 18	tests/test_us5_2_sort_filter.py
Integration	WEB-EX-04, 08, 09, 15	tests/test_stage13_integration.py

7. Key Boundaries and Pass Rules

Boundary	Pass rule
Completeness	Below 50% is Insufficient with no band; exactly 50% is eligible with caveat
Risk bands	Low 0-9 / 100; Medium 10-59 / 100; High 60-100 / 100
CSV	Maximum 100 KB in the browser; exact account_id header; 0-100 rows accepted by current contract; 101 rejected
Decision	Only completed scored result; Override reason required; duplicate prevented
Follow-up	Completed or Insufficient result; reason/status visible; update and clear
Privacy/oversight	Offline fixtures; minimal record; no platform action or definitive verdict
Return-state privacy	Only the displayed whitelisted preview/result and visible review UI state are kept for the current tab after strict schema validation; no training label or raw source payload; Start new clears it
Accessibility	Text plus colour; labelled controls; keyboard path; no clipping at 200%

A case passes only when every stated expected result is visibly true. Partial success is recorded as Fail and must not be silently marked Pass.

8. Exit Criteria and Definition of Done

This Testing Plan is ready for handover when:

- the first three selector options match the documented label-free dataset rows;
- all six tracked browser-upload files open and contain the documented rows;
- the PDF matches the latest Markdown and is readable in monochrome;
- all original automated tests, JavaScript syntax and Run Sheet checks pass;
- every WEB-EX case has a truthful status or remains explicitly Not executed;
- failures have a defect ID and retest result;
- no Critical or High defect remains unresolved; and
- the team updates Trello and approval status without inventing evidence.

Writing the case is not proof that the GUI passed it. Direct browser execution, clean-environment confirmation and target-user UAT remain separate team actions.